

Cloud Infrastructure, Distributed Systems and Containerization

Student Name: Neha Gupta | Roll Number: CS-2024-033

1. Principles of Cloud Computing

Cloud computing delivers on-demand computational capabilities including virtual servers, distributed object storage, managed database engines, and software services over high-speed networks with elastic scalability.

2. Containerization and Microservices Architecture

Containers bundle application software along with its runtime system libraries into lightweight isolated user spaces. Tools like Docker and Kubernetes enable dynamic orchestration and self-healing deployments across heterogeneous server clusters.